

Ontology of Becoming

Abstract

If a medium marks ‘no distinction is drawn’ differently from its denial, then it has already made a distinction. This paper develops this idea in four steps. First, a law that remains unchanged under any relabeling of its carrier only moves a point when the carrier has two values. In that case, the only such law besides the identity is negation. When seen as an equation, negation has no solution. When seen as a recursion, it produces a unique period-two alternation from any starting point. If a tokening can distinguish between the denial of an occurrence and its confirmation, it also confirms the occurrence itself. Each of these results holds for static configurations. The final section considers the temporal perspective: when claims are treated as properties of worlds without succession, a claim that is true wherever its denial is distinguished will also be true in worlds that distinguish it. This means that temporal realisation depends on a premise, with one instance for each realised step, which is stated and proved separately from the main development. Instantiating in the world requires one empirical input (a discriminating tokening) provided by the page. The obstruction theorem formally matches Stroud’s limit on transcendental arguments and Trendelenburg’s objection to Hegel. The main formal results are checked in Lean 4 without using any axioms. The interpretive steps that link these results to the idea of becoming are explained and justified in the text.

1 Introduction

Classical ontological arguments examine a concept and reach a conclusion about existence based on reasoning alone [3, 14, 21]. Here, the concept is simply a medium’s ability to distinguish one proposition from another. The conclusion has two parts, each with its own implications.

The first part is structural and does not rely on any assumptions. It consists of a set of proven conditionals. On any medium, a rule that depends only on the difference between values remains unchanged if the labels are switched. This rule only changes a point on a two-valued medium, where the rule is negation. Negation does not have a fixed point, and its sequence from any starting value alternates between two states, which is unique except for how the values are labeled. If you mark the denial of these facts differently from their assertion, the marking itself provides an example.

The second part deals with time, while all earlier results can be met by a static setup. Section 6 shows that the difference is structural: when claims are treated as properties of worlds without succession, a claim that is true wherever its denial is recognised also holds in static worlds that recognise it. However, static worlds do not progress, so the time-based interpretation remains possible. Making this work over time depends on a premise, called *P*: the marking of the denial and the marking of the assertion happen one after the other. One instance of *P* gives a single step; a series of instances creates a sequence; and both the single step and the sequence are shown to be independent of what was previously derived.

- (i) *Verified conditionals*. The formal results of Sections 3, 4, and 6, checked in Lean 4 with an empty axiom footprint (Section 8).
- (ii) *Regimentation choices*. Five identifications connect the concept to the formalism: ground

as law, structureless as equivariant, groundhood as exclusion, satisfaction as objectual or processual (Section 3), and claims as properties of succession-free worlds (Section 6.3).

- (iii) *Empirical fact*. The actual world contains a tokening that discriminates the relevant pair. Section 5 discharges this by ostension and counts it; the anti-bridge corollary of Section 6 repeats the discharge schematically, one tokened pair per candidate.
- (iv) *Premise P*, with its orientation, the reading of the added relation as directed succession, and, for the full run, an infinite family of instances over pairwise-distinct loci. Section 6 prices it.

Retorsive arguments have a well-known problem: even if no one can deny them, that does not guarantee they are true about the world [60]. This section proves a stronger version of that problem as a theorem for this case, and then shows that the argument's premise sits right at the boundary set by the theorem, using an independence proof in the model-theoretic style. The same theorem also formalizes a nineteenth-century objection to Hegel's derivation of becoming (Section 7).

2 The Argument

Once you have a distinction, consider what kind of rule could apply to the medium if it only uses the fact that the values are different. If there is only one value, the only possible rule is to leave it unchanged. With three or more values, any rule that ignores labels must leave everything as it is. Two values are different: in this case, a rule that does not care about labels can still change things, and that rule is negation. The identity rule applies to everything and does not highlight anything; negation is the only simple rule that can exclude something.

Negation does not have a fixed point, so nothing unchanging fits the basic idea. The sequence defined by $x_{n+1} = \neg x_n$ can start from any value, and the result is always unique and alternates every two steps. If you create a relation based only on equality and it leaves out any case where something equals itself, it leaves out all such cases. If the relation is ever true, it is true for a sequence like this.

Finally, tokening the denial of all this in a way that is clearly different from its assertion, you show a difference: two marks, two values, separated by one negation. With two values, this just repeats that they are different. The marking shows that something happens, but whether the marks come one after another is a separate question that depends on the main idea.

3 Regimentation

The concept under analysis is presuppositionlessness, or a ground that uses nothing beyond a bare distinction. This section defends the four choices that bring it into the formalism.

3.1 The ground is a law

A *carrier* is a type V ; a *law* on V is a function $f : V \rightarrow V$. Choosing to interpret the ground as an operation is just one option. Other possibilities are to see it as a relation, a constant, or simply a set. A constant assigns a value, but this idea of naming is not allowed by the concept, as explained more formally below, where constant maps do not meet the requirement of structurelessness. The relational approach is discussed in Section 4 along with the operational one, and both approaches come together: the relation used there is disagreement, which is the graph of negation, and Fix and Run of a law match up with object-based and process-based satisfaction of its graph.

3.2 Structureless means equivariant

For $a \neq b$ in V , the transposition τ_{ab} exchanges a and b and fixes the rest.

Definition 3.1 (Equivariance). $f : V \rightarrow V$ is equivariant if $f(\tau_{ab}(x)) = \tau_{ab}(f(x))$ for all $a, b, x \in V$.

Transpositions form the finitary symmetric group, and the theorems below use transposition-equivariance directly. Here, equivariance means commuting with every finitary relabeling. If a law depends only on the distinctness of values, it must commute with relabeling, since relabeling only changes which token represents which value. The reverse idea, that invariance is enough for definability from distinction alone, is known as the permutation-invariance criterion for logical notions. This is a debated topic with its own literature [61, 51, 39, 16, 8, 63]. McGee’s theorem matches invariant operations with those definable in $L_{\infty\infty}$, while Feferman’s overgeneration objection highlights the difference between invariance and innocence.

For the two-valued carrier, the two interpretations are provably the same.

Proposition 3.2 (Coincidence on two values). *On $\mathbb{B}$: the structureless unary laws are the identity and negation, whose graphs are equality and disagreement, both definable (Definition 4.12); and a Boolean-valued binary relation invariant under joint negation is extensionally equality, disagreement, the full relation, or the empty one.*

Proof. Unary: Proposition 4.7, with the graphs read off. Binary: fix $f(\top, \top)$ and $f(\top, \perp)$; invariance determines $f(\perp, \perp)$ and $f(\perp, \top)$, and the four cases give the four relations; verified as `invariant_binary_classified`. Identifying the four with the `Defble` extensions is immediate: equality, its negation, their disjunction, their conjunction. $\square$

The equivalence between invariance and definability is used only on $\mathbb{B}$, where it is a proven theorem. For carriers of any size, the development relies solely on the necessity direction, which is undisputed. This leads to one restriction for everything that follows: the formalisation of equivariance and the arena result are both stated for carriers with decidable equality.

3.3 Groundhood requires exclusion

The *ground* is a structureless law that goes beyond identity. This idea comes from the most fundamental level: the concept under discussion concerns the ability to distinguish one proposition *from* another. Under identity, nothing is distinguished from anything else, so a ground that fits this concept must exclude something. The classical phrase for this is *omnis determinatio est negatio* [57]. A Parmenidean ground that does not distinguish anything is still possible, but only if we give up the concept. This condition is made formal as `Says` in Section 4.

3.4 Satisfaction is objectual or processual

Section 4 defines satisfaction of a relation as a choice: either a value satisfies it directly, or a sequence satisfies it step by step. This choice sets the options for comparison: a state compared to itself, and a sequence compared stepwise. Other options are possible, such as satisfaction by a pair or by a set. Theorem 4.17 gives its result for relations that fit within this framework.

3.5 What retorsion is

The retorsion step is a formal point about cardinality. If a function $t : \text{Prop} \rightarrow M$ takes two different values, it shows there are two distinct elements in M , no matter which two propositions you use. When we use $\neg\text{Draws}(M)$ and $\text{Draws}(M)$ as the propositions, the example becomes more pointed, because the fact being shown is the one being denied. In Mackie’s terms, this is

called pragmatic self-refutation [37, 12]: the denial is true in some cases, but false whenever it is used in a specific way, which is what Corollary 4.11 shows. Hintikka's view of the cogito as a performance is the closest classical example [27]. Kaplan's distinction between validity in every context and necessity highlights the gap that Section 6 makes precise: a claim can be true whenever it is used, but still be contingent [31].

4 Formal Development

4.1 The arena

Proposition 4.1 (One value is mute). *If V has exactly one element, every law on V is the identity.*

Proof. Only one function $V \rightarrow V$ exists. □

Proposition 4.2 (Three or more values force triviality). *Let V have decidable equality and at least three elements. Every equivariant $f : V \rightarrow V$ satisfies $f(x) = x$ for all x .*

Proof. Suppose $f(x) \neq x$. Choose z with $z \neq x$ and $z \neq f(x)$. The transposition $\tau_{f(x),z}$ fixes x , so equivariance gives

$$f(x) = f(\tau_{f(x),z}(x)) = \tau_{f(x),z}(f(x)) = z,$$

contradicting $z \neq f(x)$. □

Proposition 4.3 (A moved point exhausts the carrier). *Let V have decidable equality, f equivariant, and $f(x) \neq x$. Then every $z \in V$ equals x or $f(x)$.*

Proof. Suppose z differs from both. As in Proposition 4.2, $\tau_{f(x),z}$ fixes x and equivariance forces $f(x) = z$, a contradiction. □

Proposition 4.4 (Two values can move). *On $\mathbb{B} = \{\top, \perp\}$, negation is equivariant and $\neg x \neq x$ for both x .*

Corollary 4.5 (The arena). *A carrier with decidable equality admits a non-trivial equivariant law only if it has exactly two elements, and $\mathbb{B}$ admits one.*

Proof. Propositions 4.1, 4.3, and 4.4. □

From here the carrier is $\mathbb{B}$. On $\mathbb{B}$ the sole non-identity transposition is the swap of the two values, which is negation, so equivariance specialises as follows.

Definition 4.6 (Structureless law on $\mathbb{B}$). *$g : \mathbb{B} \rightarrow \mathbb{B}$ is structureless if $g(\neg b) = \neg g(b)$ for every b .*

Proposition 4.7 (Uniqueness of the law). *A structureless g is id or $\neg$.*

Proof. Case on $g(\top)$; structurelessness determines $g(\perp)$. □

Proposition 4.8 (The identity says nothing). *Constant maps on $\mathbb{B}$ fail structurelessness: $g(\neg \top) = c$ and $\neg g(\top) = \neg c$ differ. The identity is structureless and fixes every element.*

With Section 3's exclusion condition, the ground, if anything is, is negation.

4.2 Retorsion

Definition 4.9 (Draws). *A medium M draws a distinction, $\text{Draws}(M)$, if there exist $a \neq b$ in M .*

Proposition 4.10 (Retorsion). *For $t : \text{Prop} \rightarrow M$: if $t(\neg \text{Draws}(M)) \neq t(\text{Draws}(M))$, then $\text{Draws}(M)$.*

Proof. The two token values witness the distinction. □

Corollary 4.11. *The conjunction of $t(\neg \text{Draws}(M)) \neq t(\text{Draws}(M))$ with $\neg \text{Draws}(M)$ is contradictory: a discriminated tokening of the denial makes the denial false.*

As Section 3.5 notes, Proposition 4.10 holds for any pair of propositions; its content is the cardinality of the medium. The self-referential instance directs that content at the denial itself.

4.3 Definable relations

Definition 4.12 (Definable relations). *The definable relations on $\mathbb{B}$ are generated from equality by negation, conjunction, and disjunction, pointwise.*

Equality comes from the basic distinction, while the connectives allow for combinations. There are no constants in the base, since a constant relation refers to a value. Proposition 3.2 shows that the grammar captures only the invariant relations.

Lemma 4.13 (Invariance). *Definable R satisfies $R(a, b) \leftrightarrow R(\neg a, \neg b)$.*

Proof. Induction on generation. □

Lemma 4.14 (Uniformity). *Definable R satisfies $R(a, a) \leftrightarrow R(b, b)$.*

Proof. The joint swap maps the diagonal onto itself; Lemma 4.13 carries truth along the orbit. □

This is a symmetry argument where an invariant relation is constant on the orbit of the diagonal. Its relationship to the fixed-point results is discussed in Section 7.

4.4 Articulate relations

Definition 4.15. *For a relation R on $\mathbb{B}$: R excludes x if $\neg R(x, x)$; $\text{Says}(R)$ if it excludes some value; R is objectually satisfied if $\exists x. R(x, x)$; processually satisfied if some $s : \mathbb{N} \rightarrow \mathbb{B}$ has $R(s(n), s(n+1))$ for all n ; $\text{Sat}(R)$ if either; R articulates if $\text{Says}(R)$ and $\text{Sat}(R)$.*

Proposition 4.16 (Exclusion closes the object frame). *Definable R with $\text{Says}(R)$ is objectually unsatisfied.*

Proof. Uniformity spreads any reflexive satisfaction to the excluded value. □

Theorem 4.17 (The law of an articulate relation). *Definable, articulate R satisfies: (a) $R(a, b) \leftrightarrow a \neq b$; (b) objectual satisfaction fails; (c) some sequence satisfies R stepwise; (d) every such sequence is determined by its first term, alternating thereafter.*

Proof. By Proposition 4.16, satisfaction is processual; the first step gives $s(0) \neq s(1)$; invariance transfers R across both orderings of the two values, giving (a); (c) is the orbit of $\neg$; (d) is induction using (a). □

4.5 No being

Theorem 4.18 (No being). $\neg x \neq x$ for every $x \in \mathbb{B}$, and $X = \neg X$ has no propositional solution.

Proof. Case check; and $X \leftrightarrow \neg X$ yields both X and $\neg X$. $\square$

4.6 The alternation

Definition 4.19 (Orbit). The orbit of f from x_0 is the sequence with $x_{n+1} = f(x_n)$.

Theorem 4.20 (Alternation). The orbit of $\neg$ from either seed solves the recursion, is the unique solution with that initial value, has period two, and the two orbits are relabelings of each other. Moreover

$$(\forall x. \neg x \neq x) \iff (\forall \text{seed } x_0, \forall n. \text{orbit}_{\neg}(x_0)(n+1) \neq \text{orbit}_{\neg}(x_0)(n)).$$

Proof. Existence is the definition; uniqueness the induction; period two is $\neg\neg x = x$; relabeling is induction. The displayed equivalence holds because the two sides quantify over the same instances: each direction is instantiation. $\square$

Remark 4.21. The equivalence is a requantification, and the absence of a fixed point and stepwise movement of every orbit are two descriptions of one hypothesis.

4.7 Discrimination

Definition 4.22 (Occurs). For $h : T \rightarrow \mathbb{B}$: $\text{Occurs}(h)$ if $h(t_1) \neq h(t_2)$ for some $t_1, t_2 \in T$.

Theorem 4.23 (Discrimination). Let $\text{tok} : \text{Prop} \rightarrow T$ and $h : T \rightarrow \mathbb{B}$. If $h(\text{tok}(\neg \text{Occurs}(h))) \neq h(\text{tok}(\text{Occurs}(h)))$, then $\text{Occurs}(h)$ and the two tokenings are distinct.

Proof. Proposition 4.10 for the first claim; equal tokenings would give equal values for the second. $\square$

Remark 4.24 (Deflation). The verified statement also includes a third part: the two values differ by one negation. With only two values, this just means they are distinct, so every pair of different marks in any binary classification has this property. The term "tick" will be used in Section 6, where it fits better because of the idea of succession.

4.8 Synthesis

Theorem 4.25 (Synthesis). The conjunction of Corollary 4.5, Propositions 4.7, 4.8, 4.10, Lemma 4.14, and Theorems 4.17, 4.18, 4.20, 4.23 holds. Each conjunct is a conditional; detachment of any consequent for the actual world uses the empirical input of Section 5.

5 Scope

Theorem 4.23 is a conditional over an arbitrary domain T . Its antecedent is an empirical matter, and this section discharges it by tokening the pair:

Occurrence fails of every reading of this page.
Occurrence holds of some reading of this page.

The two sentences are meant to express different ideas, and any way of classifying their marks gives an h that makes the first part true. The discharge happens by showing and choosing a classifier, and this is a basic assumption: in reality, there is a medium, a token, and a way to tell them apart. In the argument, the first part remains open, as explained by the metatheory: a closed normal-form term of type $\text{Prop} \rightarrow \text{Bool}$ cannot use its propositional argument, so it

stays constant, which is a well-known result from normalisation in the axiom-free calculus (for details, see [11]). Detachment takes place outside the system. This is a metatheorem and is cited as such.

The result of the discharge is precise. In this case, the world has two different marks based on the chosen classification. Remark 4.24 applies here: for two values, being ‘one negation apart’ is the same as being ‘distinct.’

Every theorem so far is satisfied by a static configuration. Take T to be the marks on a printed page and h any separating classification; the hypothesis and conclusions of Theorem 4.23 hold for that motionless instance. The formal alternation is likewise an ω -indexed sequence of values, a tenselessly specifiable object. Realisation in temporal succession is a further matter which will be taken up in the next section.

6 The Obstruction and Premise

6.1 Worlds and claims

A *world* is a triple $W = (T, h, \text{tok})$: loci, a two-valued reading, a tokening. A *claim* is a property of worlds. W *discriminates the denial* of claim A when $h(\text{tok}(\neg A(W))) \neq h(\text{tok}(A(W)))$. A is *undeniable* when it holds at every world that discriminates its denial. The notion is inhabited: occurrence is undeniable, by Theorem 4.23 transposed to worlds.

A *tensed world* adds a succession relation on the loci. It is *still* when nothing succeeds anything; it *runs* when an infinite sequence of pairwise-distinct loci, each succeeding the last, carries alternating values. Distinctness of the loci belongs to the definition: a two-locus cycle satisfies the succession and alternation clauses alone, and counts as a run for nobody.

6.2 Freeze and obstruction

Proposition 6.1 (Freeze). *Two tensed structures over one world satisfy exactly the same claims. Over any inhabited world, a still structure and a running-capable one coexist, and every claim holds at both or at neither.*

Proof. Claims are typed over worlds, so the invariance is definitional; the verified proof is reflexivity. For the second half, equip the world with the empty succession and with the total one. $\square$

Theorem 6.2 (Obstruction). *Let A be undeniable and entail running. Then every still tensed world fails to discriminate the denial of A .*

Proof. A still discriminating world would satisfy A , hence run, and stillness refutes running at the first step. $\square$

Corollary 6.3 (No undeniable bridge). *Given a still tensed world that discriminates the denial of A , an undeniable A fails to entail running.*

The corollary’s hypothesis is handled in the same way as in Section 5: the printed page, seen as a fixed set of marks, represents a world where denials can be distinguished. The process is schematic, like P : the page must show both the denial and assertion for each candidate A , one pair for each, covering all candidates the proponent mentions. Therefore, an axiom that could close the gap would need to be both undeniable and capable of connecting the two, but these qualities cannot coexist on the page.

6.3 The regimentation

According to the performative view, a tokening is an act, acts are events, and the facts that retorsion secures are tensed [4, 27]. This proposal formally assigns claims to tensed worlds. However, if the tensed-level discrimination relies only on the world components, the obstruction argument repeats at the tensed level: a static world whose world part distinguishes the denial produces the claim, and the lack of change prevents progress; this version is called *obstruction_tensed*. If the notion instead uses the succession component, then having a discriminating act assumes a succession fact, so tensed content appears along with succession, valued as P is. Either way, the situation remains the same. This dilemma supports the regimentation, and the obstruction shows where undeniability can exist, given that tokening facts are limited to the world triple. Kaplan’s idea of the contingency of the context-valid marks the same boundary in different terms [31].

6.4 The premise

Let the premise be:

P : the tokening of the denial and the tokening of the assertion stand in succession.

By Proposition 6.1, P ’s denial is realized by a still structure over the very same world, so P is deniable; by Corollary 6.3, an undeniable replacement is unavailable. Its standing is Moorean [42]. Given one instance, one step of the alternation is realized in succession: the successor’s value is the negation of its predecessor’s, from the discrimination hypothesis, as *actuality_tick* proves. This premise-conditional result is the one the paper calls *actuality*. An infinite chain of instances over pairwise-distinct loci, pairwise discriminated, realizes the run in the full tensed sense, and the formal *Runs* requires exactly this: distinct loci, succession, alternation. The run therefore costs one instance of P per step, and the count is the schema with its infinite family, stated as such.

Proposition 6.4 (Independence). *Over one world carrying two distinctly read loci: the still structure satisfies every derived consequent of Theorem 4.23 wherever discrimination holds, and P fails for every tokening; the running structure satisfies P for every tokening and the succession half of the chain hypothesis for every sequence. The full chain hypothesis, distinct loci and discrimination included, is satisfied in a world with infinitely many loci (*chain_hypothesis_realizable*). The single instance and the chain hypothesis are each underivable and irrefutable from the development.*

Proof. Empty succession and total succession over the same two-locus world; Proposition 6.1 gives that the structures share all tokening facts. The still witness has a constant tokening, so discrimination holds there only for hypothetical tokenings; the quantification is over all of them. For the full chain hypothesis, alternate a reading over $\mathbb{N}$ loci under total succession. $\square$

P , as formalized, puts denial before assertion. The alternation itself does not have a built-in direction, since predecessor values exist and are required (Proposition 6.5), and extending backward is similar to extending forward. This means the chain’s direction is set by the premise and recorded with it. It is important to be precise when naming the refuser of P . The tensed world adds a relation that is always read as directed succession. Someone who rejects this entirely only keeps an undirected order among points, which is the C-theorist’s view [15]. The eternalist accepts tenseless succession [52], so they accept P when read without tense. In that view, a chain of instances shows variation along a tenseless order, and the idea of passage, as the A-theorist sees it, is a further step beyond what this paper covers [66, 40]. So, there are two gaps: from structure to succession, which P addresses, and from succession to passage, which remains open. In this paper, “becoming” refers to the alternation and its realization in succession. Readers who use “becoming” only for passage should adjust the term as needed.

6.5 Rigidity

Proposition 6.5 (Rigidity). *At the level of values: (a) a token alternating with a tick and with its successor forces $x = \neg x$, so a realized alternation admits no alternation-refining interpolation; (b) a predecessor value always exists and is forced, so any two candidate predecessors of a tick agree in value. With forward uniqueness (Theorem 4.20), one realized tick determines the values of every tick before and after it, a closure assembled in prose from *becoming_unique* and *prior_token_determined*.*

Proof. (a) $h(u) = \neg h(t)$ and $h(\text{succ}(t)) = \neg h(u)$ with $h(\text{succ}(t)) = \neg h(t)$ force $h(t) = \neg h(t)$.
(b) $\neg h(t)$ is the unique candidate. $\square$

The division of labor is now complete in both directions. Existence of realized ticks is premise, instance by instance. The values of any realization are theorem, rigid throughout.

7 Related Work

7.1 Ontological arguments

Anselm, Descartes, and Gödel argue that necessary existence follows from a concept [3, 14, 21, 45]. Oppy reviews this type of argument [44]. In this paper, I use a similar structure but change both terms: here, the concept is the ability to make a distinction, and what follows is alternation, not a fixed answer. Findlay’s ontological disproof is the closest formal ancestor to Theorem 4.18 [18], and Hartshorne developed an ontological argument within process metaphysics [24]. Leibniz insisted on the possibility premise behind Anselm and Descartes [36]. My approach is to prove everything that can be proved and to clearly identify the one independent premise, following Leibniz’s demand with modern methods. Kant objected to arguments that include worldly existence in a concept [30]. My response is clear: the a priori part of this work gives mathematical existence, while real-world content comes in with the empirical step and with *P*. Theorem 6.2 shows, in this context, a version of Kant’s idea that every existential claim is synthetic. So, the obstruction supports Kant’s view.

7.2 Verified ontological arguments

Oppenheimer and Zalta made Anselm’s argument simpler by using an automated reasoner [43]. Benz Müller and Woltzenlogel Paleo checked Gödel’s proof with higher-order provers, confirming the modal collapse of the Scott variant [6], which philosophers have known about since Sobel [55]. They later showed that Gödel’s original axioms are inconsistent [7]. Anderson’s changes balance the strength of the axioms with the strength of the conclusions [2]. These cases put existential content into the premises and let machines check the cost. In contrast, the current approach does not use any axioms and instead pays the price elsewhere: its existence claims are mathematical, and its real-world and time-related content is discussed in Sections 5 and 6. Conditionalisation shifts where the content appears, but both ends of the trade-off keep the same amount of existential import.

7.3 Retorsion and transcendental arguments

Aristotle’s elenctic defence of non-contradiction is the earliest example of retorsion [5]. Apel and Habermas turn performative contradiction into a foundational approach [4, 23]. Stroud argues that transcendental arguments only reveal facts about those who deny them, not about the world itself [60]. Stern distinguishes between modest and ambitious versions of these arguments [58], and Illies reviews the broader retorsive tradition [29]. This paper’s contribution is a theorem that proves the modest result that occurrence is shown wherever denial is identified, blocks tensed

realisation, and marks the boundary with an independence proof. In the *Letztbegründung* debate, this view supports Albert over Apel for the temporal case [1], using the dilemma in Section 6.3. Strawson's later naturalism, which focuses on what deniers must accept, is the informal basis for the premise's Moorean status [59]. Mackie, Hintikka, and Kaplan provide the classification used in Section 3.5 [37, 27, 31].

7.4 Laws of Form

Spencer-Brown starts by asking us to draw a distinction. In Chapter 11, he resolves an unsatisfied equation by introducing an oscillation, which he describes as the introduction of time [56]. Kauffman and Varela expand on this idea of oscillation [33, 32, 64]. Their framework anticipates the approach in this paper, including the focus on time, and I acknowledge these influences while pointing out important differences. *Laws of Form* takes its two-valued arithmetic as a given, while in this paper, the two-valued setting and the uniqueness of negation are proven results (Corollary 4.5, Proposition 4.7). In *Laws of Form*, oscillation is introduced through a specific construction, but here it arises from classifying articulate relations (Theorem 4.17). The temporal interpretation is stated simply in their work, but in this paper, Theorem 6.2 shows that this interpretation cannot be derived from unquestionable principles and assigns it the value P .

7.5 Hegel and his critics

Hegel uses the term presuppositionlessness to describe the start of his *Logic*, where pure Being cannot stand on its own and turns into Becoming [25]. Houlgate and Maker discuss what can come from such a beginning [28, 38]. In this paper, the main argument follows Hegel's approach, but uses combinatorics instead of dialectic. The main challenge comes from Trendelenburg, who argued that pure thought cannot create movement and that dialectic borrows motion from intuition [62]. This idea is also found in Schelling's claim that existence must go beyond negative philosophy [50], and in Kierkegaard's criticism of movement in logic [34]. Here, this challenge is treated as a general rule for the field. The only extra assumption is what Schelling said the positive must add. With this approach, the paper shows that both sides contribute: structure comes without extra cost, but movement needs one new premise at each step. Some, like Priest and Ficara's reading of Hegel, see the opening differently. They argue that being and nothing form a real contradiction, and they reject the explosion in Theorem 4.18 and the need for consistency [46, 17]. This paper uses a constructive and consistent logic, and the earlier conclusions depend on that choice.

7.6 Temporal ontology

McTaggart separated the C-series (ordered but not directed) from the B- and A-series [40]. The world versus tensed-world distinction is broader, since it adds a simple directed relation to an unordered base, and the C-theory rejects that relation entirely [15]. The formal alternation here is at-at variation, which Williams, Smart, and Grunbaum contrast with the idea of passage [66, 54, 22]. This is why Section 6 identifies a second gap. Broad and Prior highlight what someone who believes in real becoming would want beyond this [9, 47]. Grunbaum argues that becoming starts with the act of tokening, which is almost the opposite of the current approach, where all content is stored in tokening facts and succession is treated separately. Mellor's token-reflexive approach is the closest formal match to the tok/h system: it gives tenseless truth conditions for tensed tokens, with tokening facts doing the main work [41]. Proposition 6.1 fits well with this. Fine's fragmentalism does not fit within the still/running distinction and shows a limit of the current framework [19]. Skow's work on the nature of becoming helps define the second gap [53], and Putnam's relativistic argument is relevant to any claim about frame-independent succession [48]. In the process tradition [65, 49, 26], there is one clear point

of contact: Proposition 6.5(a), which excludes alternation-refining interpolation, is similar in scope to Whitehead’s epochal theory of becoming.

7.7 Fixed points

The main idea behind this development is the fixed-point-freeness of an involution (Theorem 4.18). Lawvere’s fixed-point theorem brings together the classical diagonal arguments [35, 10, 20], and Yanofsky’s survey gives an overview of the field [67]. In Lawvere’s contrapositive form, a fixed-point-free map leads to a diagonal result. Lemma 4.14 is itself a symmetry argument involving two elements. The underlying mechanism is related to the classical results, which are mentioned for context.

8 Verification

Every labeled result of Sections 3, 4, and 6 is a named declaration in one of two standalone Lean 4 files [13], prelude only, with no sorry, no axiom, and no import. Each file ends with a `#print axioms audit` of its main results; every line reports independence from all axioms, Lean’s standard three included. Two calibrations belong with that report. The development is constructive over types with decidable equality, and its proofs are finite case analysis and induction; for such material the empty axiom footprint is the expected outcome, and the audit records it. And the kernel checks the conditionals: the regimentation of Section 3, the empirical discharge of Section 5, and the standing of P are argued in prose, with Section 5’s canonicity observation showing the discharge lives outside the system as a matter of metatheory.

Here	Lean
Proposition 3.2	<code>one_candidate, invariant_binary_classified</code>
Corollary 4.5	<code>bare_one_is_mute, bare_three_is_mute, two_at_most, two_speaks</code>
Propositions 4.7, 4.8	<code>one_candidate, constants_not_structureless, id_says_nothing, id_structureless</code>
Proposition 4.10, Corollary 4.11	<code>retorsion, no_undrawn_denial</code>
Lemmas 4.13, 4.14	<code>defble_invariant, diagonal_uniform</code>
Proposition 4.16	<code>says_kills_the_object</code>
Theorem 4.17	<code>absolute_law, absolute_articulation, absolute_is_inhabited</code>
Theorem 4.18	<code>no_static_instance, no_being, no_fixpoint, ne_eq_not</code>
Theorem 4.20	<code>becoming_exists, becoming_unique, two_tick_clock, clock_runs, run_unique, refusal_is_running, seed_is_relabel</code>
Theorem 4.23	<code>event_retorsion, discrimination</code>
Theorem 4.25	<code>ontological_argument</code>
Proposition 6.1	<code>freeze, two_models</code>
Theorem 6.2	<code>obstruction, with_still_never_runs, occurs_undeniable; tensed typing as obstruction_tensed</code>
Corollary 6.3	<code>no_undeniable_bridge</code>
P , actuality, the chain	<code>Successive, actuality_tick, chain_runs</code>
Proposition 6.4	<code>premise_underivable, premise_unrefutable, premise_independent, chain_premise_independent, chain_hypothesis_realizable</code>
Proposition 6.5	<code>no_half_tick, no_interpolation, prior_value_exists, prior_token_determined</code>

Declaration names `two_tick_clock`, `clock_runs`, and `refusal_is_running` are mnemonics; their statements are atemporal facts about the orbit. `transp` is defined on all pairs, degenerate ones giving the identity, and the extra equivariance instances are vacuous. The audits comprise 30 lines in the first file and 17 in the second.

The files accompany the paper as `lean/Becoming.lean` and `lean/Obstruction.lean`, with the

toolchain pinned; an anonymized archive is supplied for review.

9 Conclusion

If a God is posited, he is not one, but becoming one.

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